

Question 1

The rate of chemical reactions is influenced by surface area.

The diagram shows three dishes, 1, 2 and 3. Each dish contains 2 g of marble chips.



- (c) (i) Name an instrument that could be used to measure 2 g of marble chips.

Electric balance

- (ii) Which dish contains marble chips with the largest total surface area? Put a tick (✓) in the correct box.

Dish 1 ☐

Dish 2 ☒

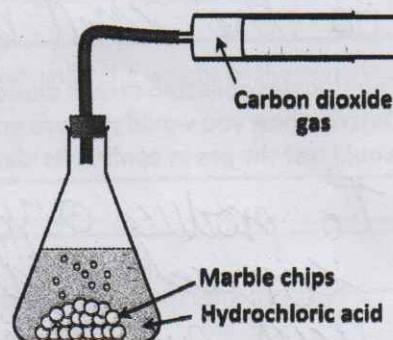
Dish 3 ☐

Marble chips react with hydrochloric acid to produce carbon dioxide gas.

A group of students was asked to investigate the effect of marble chip size on the rate of this reaction.

They used the apparatus shown on the right.

Some of the variables that might change or be kept constant during this investigation are listed below.



The concentration of hydrochloric acid	The size of the marble chips
The volume of gas produced each minute	The loudness of the noise in the room
The temperature of the acid ✓	The total mass of the marble chips ✓

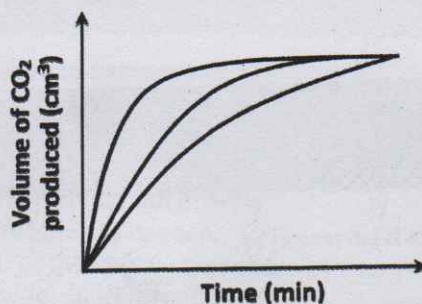
- (d) (i) From the list, identify three variables which the students should have kept constant during the investigation.

- The total mass of the marble chips
- The temperature of the acid
- The concentration of hydrochloric acid

- (ii) From the list, identify one relevant variable which changed during the investigation.

The size of the marble chips

- (e) The graph below shows the results obtained by the students.



- (i) Which of the curves (green or red or blue) on the graph corresponds to the reaction with the slowest rate? Justify your answer.

Blue because it took the longest in time (min) to reach the level that the other lines had already reached

- (ii) Which of the curves (green or red or blue) on the graph corresponds to the reaction which finished first? Justify your answer.

Green because it reached the final height on the chart before the other two

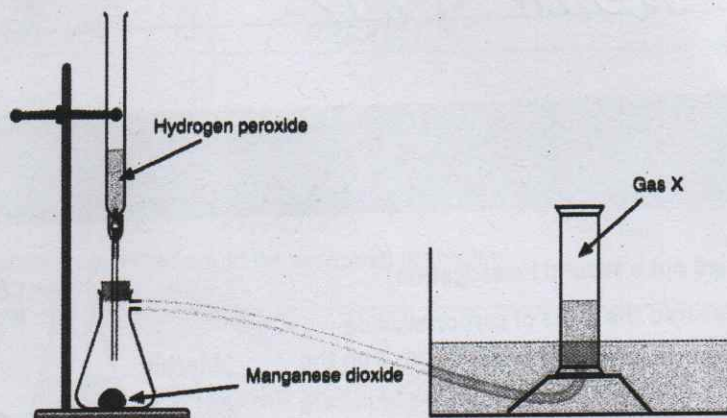
- (f) The students collected carbon dioxide gas in this investigation. Describe how you would produce another common gas in the laboratory, including how you would test the gas to confirm its identity.

To produce oxygen, you get a conical flask and put hydrochloric acid and manganese dioxide in it. You connect tubing from the conical flask through a basin filled with water and into a graduated cylinder turned upside down. To test for oxygen light a splint on fire, blow it out, but leave it glowing. Put it in the oxygen and it will relight

Question 13

(35 marks)

A student used the following set-up when studying the effect of a catalyst on chemical reactions. Liquid hydrogen peroxide was dropped onto a known amount of manganese dioxide and a reaction was observed. The gas produced was collected in a gas jar. Answer the questions that follow.



- (a) Name the piece of equipment that could be used to measure the weight of the manganese dioxide.

Electric balance

- (b) Gas X is essential for many biological processes, including respiration. What is gas X?

Oxygen

- (c) Explain what is meant by the term 'catalyst'.

A substance put into another to change the rate of reaction.

- (d) How would we observe that the gas X was being produced?

Set a wooden splint on fire, blow it out, but leave it glowing. When put in gas X, it will relight.

- (e) Would changing the amount of catalyst used during the reaction have an effect on the production of gas X? Explain your answer.

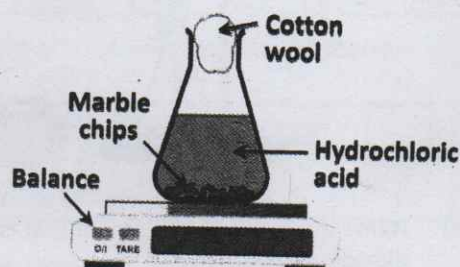
No because the catalyst itself isn't reacting so the amount of it doesn't matter

Question 3

The students carried out a second investigation.

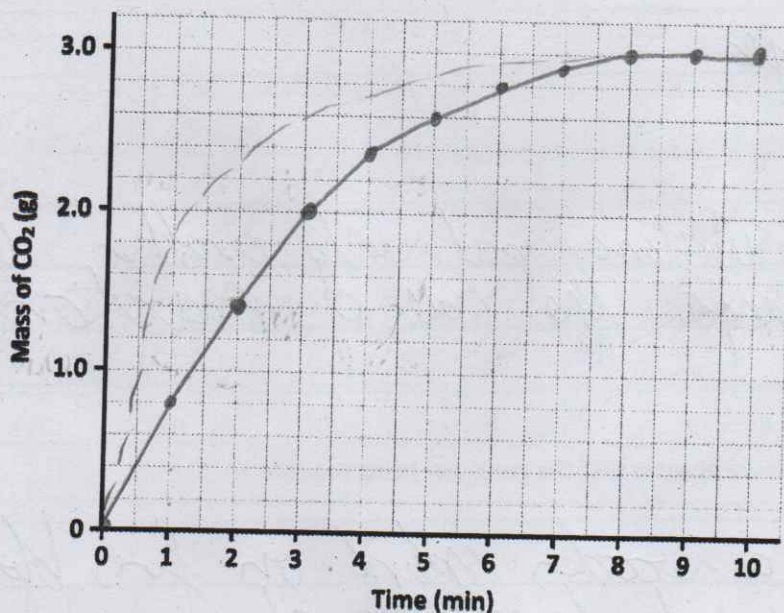
This time they measured the mass of carbon dioxide produced at regular intervals using the apparatus on the right.

The data collected by the students are shown below.



Time (min)	0	1	2	3	4	5	6	7	8	9	10
Mass of CO ₂ (g)	0	0.8	1.4	2.0	2.4	2.6	2.8	2.9	3.0	3.0	3.0

- (d) In the space below, draw a line graph of the students' results.



- (e) At what time was the reaction complete?

8 minutes

- (f) On your graph above, draw a dashed line (-----) to show the result you would expect if the above investigation was repeated exactly as before, but using smaller marble chips of the same total mass.

Question 4

- (d) Particle size is one factor that affects the rate of a reaction. Name two other factors that affect the rate of a reaction.

temperature
Nature of reactants

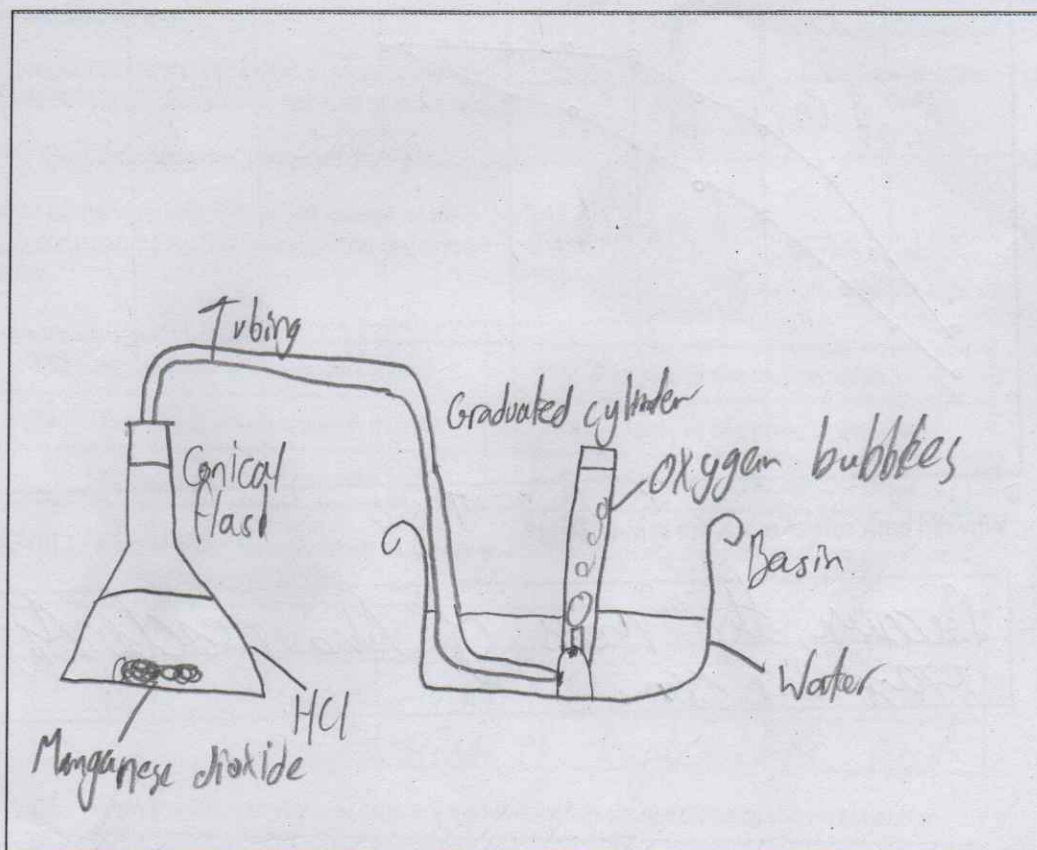
- (e) Define catalyst.

A substance added to a reaction to change its rate

- (f) Name a common gas that could be produced in the lab.

Oxygen

- (g) Draw a labelled diagram of how this gas could be produced. Include labels for any equipment and chemicals used.

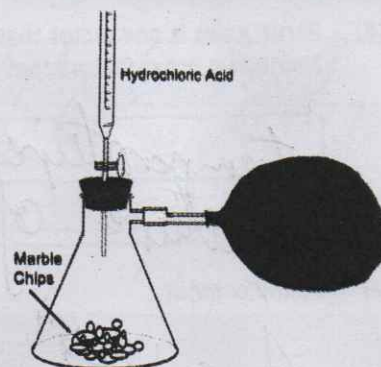


- (h) Explain how you tested this gas to confirm its identity. Include the result of the test.

How tested: Put in glowing splint
Result: Splint relights

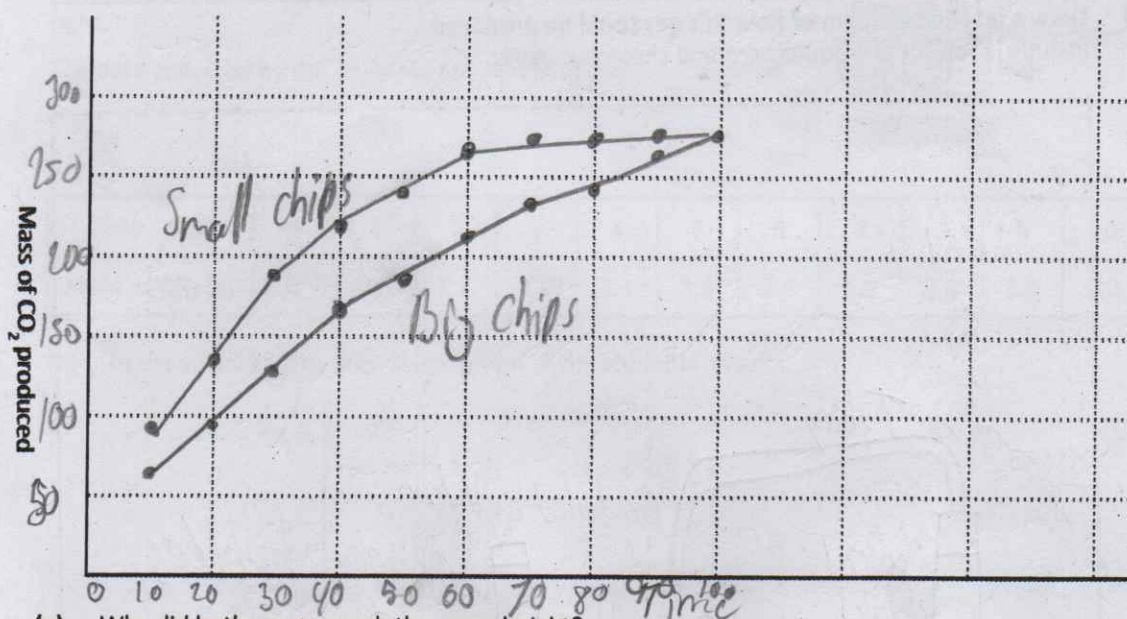
Question 5

- (f) The teacher of a class wished to demonstrate the effect that particle size had on the rate of reaction. In one instant of the experiment, the teacher added 50 cm³ of a high-strength Hydrochloric acid to some large pieces of Calcium Carbonate (marble chips). They allowed the reaction to happen, while measuring the mass of the balloon that was gathering the Carbon Dioxide. In the second instant, the teacher repeated the experiment with the same volume of acid and the same mass of marble chips, but this time the marble chips were smaller.



	10s	20s	30s	40s	50s	60s	70s	80s	90s	100s
Big Chips	62g	98g	135g	163g	189g	209g	232g	247g	257g	262g
Small Chips	96g	141g	178g	221g	245g	257g	262g	262g	262g	262g

Use the data above to draw a curve for the reaction rate for the different sized chips.



- (g) Why did both curves reach the same height?

Because the mass of the reactants was the same